K. J. Somaiya School of Engineering

Batch: A2 Roll No.: 160103240[redacted] 44
Experiment ~~/ assignment / tutorial~~ No. 06
Grade: AA / AB / BB / BC / CC / CD /DD

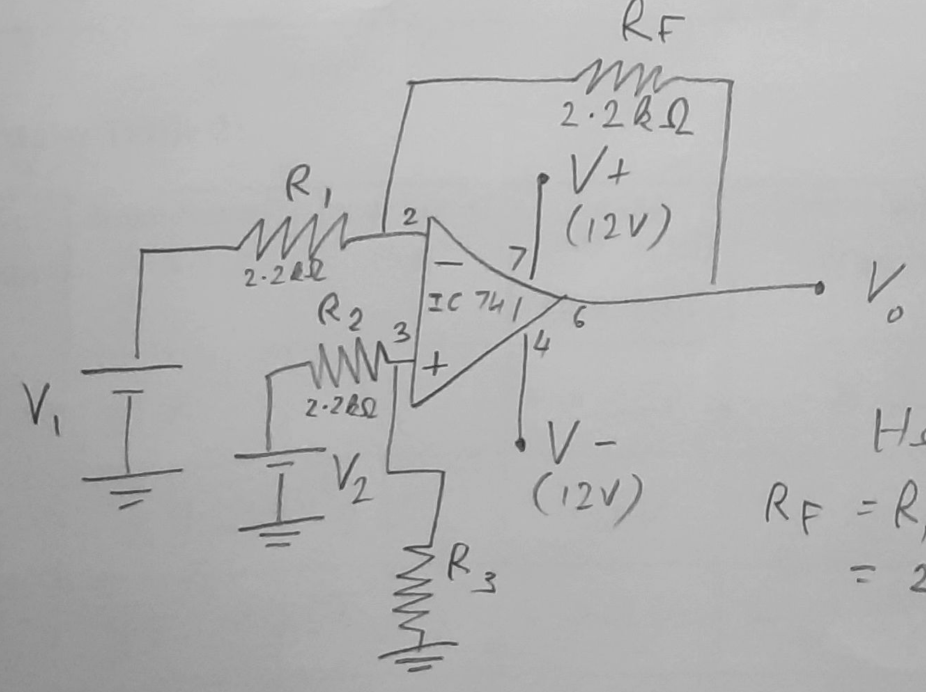
Signature of the Staff In-charge with date

TITLE: OP-AMP Linear Applications
AIM: To analyse and design adder circuits using OP-AMP
OUTCOME: Students will be able to understand the basics of analog Integrated circuits.

PROCEDURE:

1. Assemble OPAMP circuits as shown in diagram 1.
2. Analyse the circuit with reference to observation table I.
3. Apply two DC signals of different amplitudes and find the output voltage and comment.
4. In the second part, design an adder circuit as per the given input output relationship.
5. Repeat step 2 (use observation table II) and step 3 for the designed circuit.

Circuit Diagram 1: Op Amp as Subtractor



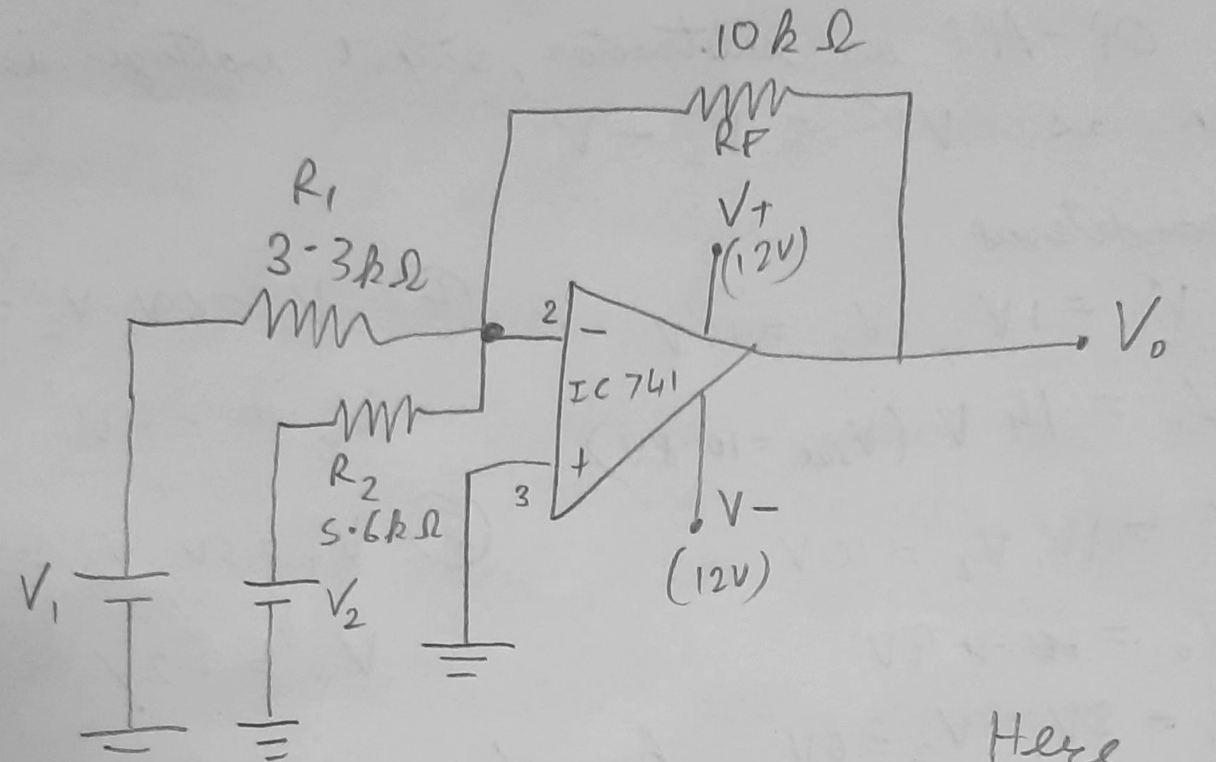
Here,
$R_F = R_1 = R_2 = R_3$
$= 2.2k\Omega$

Observation Table 1:

Sr. No	Input Signal (V_1)	Input Signal (V_2)	Output Voltage (Vo) (Practical)	Output Voltage (Vo') (Theoretical)
1.	1V	15V	11.1V (Vsat)	14V
2.	1V	10V	8.9V	9V
3.	2V	6V	4.01V	4V
4.	6V	2V	-4.015V	-4V
5.	5V	3V	-2.02V	-2V

Circuit Diagram 2: Op Amp as Adder

➢ **Design circuit for Vo = ~~-(2V1+V2)~~** $-(3V_1 + 2V_2)$



Here,

$\frac{R_F}{R_1} = 3$

$\frac{R_F}{R_2} = 2$

Observation Table 2:

Sr. No	Input Signal (V_1)	Input Signal (V_2)	Output Voltage (Vo) (Practical)	Output Voltage (Vo') (Theoretical)
1.	1V	1V	−4.95V	−4.78V
2.	2V	2V	−9.8V	−9.56V
3.	1V	2V	−7.9V	−7.78V
4.	1V	3V	−9.3V	−8.84V
5.	–	–	–	–

Theoretical Calculations:

I) For Op Amp as Subtractor:

→ For OP-AMP as subtractor, output voltage is given as $V_0' = V_2 - V_1$

For conditions

① $V_1 = 1V$, $V_2 = 15V$

$V_0 = 14\,V$ ($V_{sat} = 10.8V$)

② $V_1 = 1V$, $V_2 = 10V$

$V_0 = 9V$

③ $V_1 = 2V$, $V_2 = 6V$

$V_0 = 4V$

④ $V_1 = 6V$, $V_2 = 2V$

$V_0 = -4V$

⑤ $V_1 = 5V$, $V_2 = 3V$

$V_0 = -2V$

here, $V_{sat} = \pm 90\%\, V_{cc}$

$= \pm \frac{90}{100} \times 12 = 10.8V$

II) For Op Amp as Adder:

→ We have designed for following cond^n:-

$V_0' = -(3V_1 + 2V_2)$

Comparing with eq^n:- $V_0 = -\left[\frac{R_F}{R_1} V_1 + \frac{R_F}{R_2} V_2\right]$

$R_1 = 3.3k\Omega$

$R_2 = 5.6\Omega$ (As per lab)

→ $\frac{R_F}{R_1} = 3$ | $\frac{R_F}{R_2} = 2$

∴ Take $R_F = 10k\Omega$

For Cond^n to satisfy

$V_1 = 1V$, $V_2 = 1V$

$V_0 = (-3\times1 + 5.6\times1)$

$= -4.78$

$V_1 = 1V$, $V_2 = 3V$

$V_0 = -(3\times1 + 5.6\times3)$

$= -8.89V$

$V_1 = 1V$, $V_2 = 2V$

$V_0 = -(3\times1 + 5.6\times2)$

$= -7.78V$

$V_1 = 2V$, $V_2 = 2V$

$V_0 = -(3\times2 + 5.6\times2)$

$= -9.56V$

Conclusion: So by this experiment, We were able to say that using IC-741 OP-AMP correctly performs ~~to~~ process like addition and subtraction.

As ~~a~~ subtraction, We used 2.2kΩ

For addition we were given the equation and we used same resistor by choice and carried out the operation.

The results were near correct